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Alzheimer's Disease Detection with Deep Learning

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December 12, 2025

Contents

1	Introduction	2
2	Network Design and Implementation	3
2.1	Alzheimer Dataset Setup	3
2.2	Hyperparameter Optimization	5
3	Advanced Task	6
3.1	Data Augmentation	6
3.2	Transfer Learning	6
3.3	Grad-CAM	7
4	Neural Network Evaluation	8
4.1	Baseline Model Results and Performance	8
4.2	Improved Model Results and Performance	9
4.3	Transfer Learning Model Results and Performance	10
5	Conclusion	11
	Bibliography	12

Chapter 1

Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder characterized by memory loss, cognitive decline, and behavioural impairment. It is the most common cause of dementia worldwide and represents a major public health challenge, particularly as global life expectancy increases. According to recent estimates, over 55 million people live with dementia globally, and Alzheimer’s disease accounts for approximately 60–70% of these cases (World Health Organization, 2021) [2]. Early detection is crucial, as timely diagnosis can improve patient management, slow disease progression through therapeutic interventions, and support long-term care planning.

Magnetic Resonance Imaging (MRI) is one of the most widely used non-invasive diagnostic tools for detecting structural brain changes associated with Alzheimer’s disease.

In recent years, convolutional neural networks (CNNs) have demonstrated state-of-the-art performance in medical image classification tasks, including AD detection. For example, Suk and Shen (2014) introduced a deep ensemble model for Alzheimer’s classification using MRI and PET data, achieving significant improvements over traditional machine learning approaches [4]. Sarraf and Tofghi (2016) applied a modified LeNet and AlexNet architecture on functional MRI data, showing that deep CNNs can distinguish between Alzheimer’s and healthy controls with high accuracy [3]. More recent work by Basaia et al. (2019) demonstrated that deep CNN models trained on structural MRI can accurately classify AD, Mild Cognitive Impairment (MCI), and healthy subjects, reinforcing the potential of deep learning for early diagnosis. [1]

This project aims to explore the application of deep learning techniques for classifying Alzheimer’s disease stages using structural MRI images. Using the provided GitHub framework and publicly available datasets, we implement and evaluate convolutional neural network models capable of distinguishing among four categories: Non-Demented, Mild Demented, Moderate Demented, and Severe Demented. The baseline model is extended by experimenting with alternative architectures, including transfer learning approaches such as VGG16, ResNet, and EfficientNet, as well as data augmentation and hyperparameter optimization strategies. The main objective is to assess model performance and investigate methods that can further improve classification accuracy and robustness ¹.

¹The introduction was generated by an AI

Chapter 2

Network Design and Implementation

2.1 Alzheimer Dataset Setup

For the dataset setup, the dataset [5] was downloaded and prepared using strategies to increase the effectiveness of the training. The available dataset was highly unbalanced, with the Moderate Dementia Class having a significantly smaller number of images compared to the other classes, as can be seen in the class distribution below:

Table 2.1: Dataset distribution for Alzheimer diagnostic classes

Diagnostic class	Number of patients	Number of images
Mild Dementia	28	896
Moderate Dementia	2	64
Non Dementia	100	3200
Very Mild Dementia	70	2240
Total patients	200	6400

Due to this unequal distribution, three different techniques were chosen to mitigate the negative effects on training: Stratified K-Fold, weighted classes, and a weighted sampler for the data loader. During implementation, due to the satisfying results obtained with the combination of K-Fold and weighted classes, the weighted sampler implementation was excluded.

The K-Fold strategy ensures that every fold maintains the same ratio of classes between the training and validation sets as the full dataset. This strategy helps create a fairer division between training and validation, based on the number of folds selected (3 in this project's case); this is especially helpful in the case of unbalanced datasets. The weighted class strategy causes classes with less data to have a higher impact during the weight setting compared to the classes that are more abundant due to the assignment of higher loss weights to underrepresented classes; in this way, the data loading can lead to better training

For efficient data loading, we defined the class `AlzheimerDataModule` using the `Lightning Data Module`. The `train_dataloader` and `val_dataloader` methods were added and combined with the K-Fold strategy. In this way, by calling the class `AlzheimerDataModule`, the original dataset is split into training and validation sets applying the Stratified K-Fold, where K was selected as 3.

As input to CNN, the Alzheimer images had to be resized to match the input layer. The initial images had dimensions of (176,208), they were resized to (128,128).

For the model architecture definition, a class was declared using the Lightning Module, which allows us to describe the model's architecture, the training step, the validation step, and the performance metrics through predefined methods. A simple Convolutional Neural Network (CNN) was chosen as a baseline model to predict the class confidence values. A diagram of the designed neural network can be seen below.

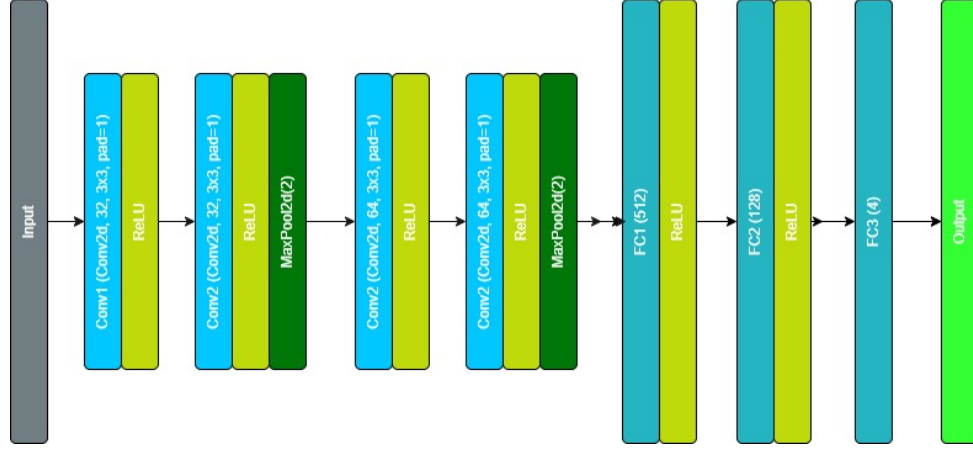


Figure 2.1: Baseline model architecture

After training, the final evaluation is performed by loading the best model previously saved and then calling the trainer in "validate" mode. The evaluation metrics selected were: Training accuracy, F1 Score, validation accuracy, validation precision, and validation recall.

In order to improve the baseline model, several modifications were made, the added strategies are described below:

- Changing the number of feature map channels in each convolutional layers;
- Fine tuning the hyperparameters using WandB (Weights and Biases) sweeps;
- Running Grad-CAM to see if the model was really learning the important Alzheimer features or just memorizing data;
- Using a second augmented Alzheimer dataset and adding salt and pepper noise on it to increase data variability. [5]

The image below shows the selected improved model.

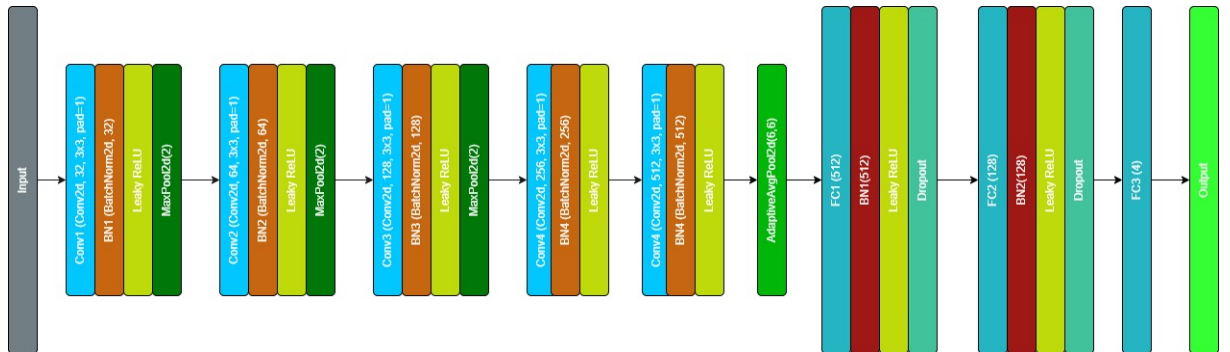


Figure 2.2: Improved CNN architecture

2.2 Hyperparameter Optimization

Hyperparameter tuning is a crucial process for maximizing model performance. Typically, this is performed on a dedicated validation set; however, as noted in the previous section, the available data is insufficient for a standard training, validation, and test split. Consequently, hyperparameter optimization was conducted on the validation set of the first fold.

To enhance training efficiency, WandB sweeps were used. This technique automates the search for the best model configuration by maximizing the accuracy of the validation. The hyperparameters evaluated during the sweeps were: neurons in the first fully connected layer, learning rate, dropout rate and batch size.

Given the high computational cost of the K-fold strategy, the sweep was restricted to a single fold (specifically, fold 0). The optimal hyperparameters identified during this phase were subsequently applied to train and validate the model across all folds.

Chapter 3

Advanced Task

3.1 Data Augmentation

The data augmentation chosen was salt and pepper noise to increase the robustness of the Alzheimer Dataset images. The figure below shows the effect of this particular noise applied to a random image taken from the augmented dataset.[5]

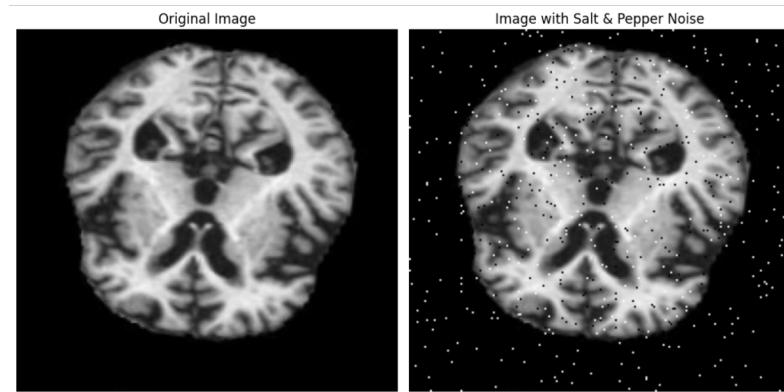


Figure 3.1: Dataset image without and with the salt and pepper noise

3.2 Transfer Learning

For Transfer Learning, **ResNet18** was selected. Initially, the network's backbone weights were frozen, and 6 sweeps were executed to analyze the hyperparameters. However, the results were suboptimal: the obtained validation accuracy was 0.74, while the training accuracy was 0.67. This indicates that the network was underfitting. Consequently, another round of sweeps was run with the backbone weights unfrozen. This yielded a significantly higher accuracy of 0.94; however, in this configuration, the network exhibited an uncertain behavior looking at the Grad-CAM result, as detailed in Section 4.3.

The low accuracy is probably due to the Alzheimer classification problem being tackled in this project being very specific, this way, the neural network employed has a very specific set of weights to analyse the types of dementia. In the case of the ResNet18 network, it was trained to identify more varied images, also looking at texture and colour, to differentiate a flower from an animal for example.

3.3 Grad-CAM

To visualize the regions of the MRI images that contribute the most to the model’s prediction, Grad-CAM (Gradient-weighted Class Activation Mapping) was employed. It consists of computing the gradient of the target class score with respect to the activations of a selected convolutional layer.

The final result is a heatmap which is an interpretable visualization showing where the network focuses during classification. The heatmap color spectrum plays an important role, it serves as a direct indicator of feature importance: red areas highlight the regions that contribute the most significantly to the correct classification, while blue areas represent the opposite or minimal contribution, with the colors in between reflecting a proportional degree of influence. This provides an essential qualitative check that the model attends to clinically meaningful brain regions when assessing dementia severity.

Chapter 4

Neural Network Evaluation

4.1 Baseline Model Results and Performance

The baseline model gave the following result:

Validate metric	DataLoader 0
val_acc	0.9714151620864868
val_f1	0.7199473977088928
val_loss	0.08917168527841568
val_prec	0.7275851368904114
val_rec	0.7128487825393677

Figure 4.1: Metrics of the baseline model

It seemed to yield a good accuracy however after trying the Grad-CAM on this network the following result was obtained:

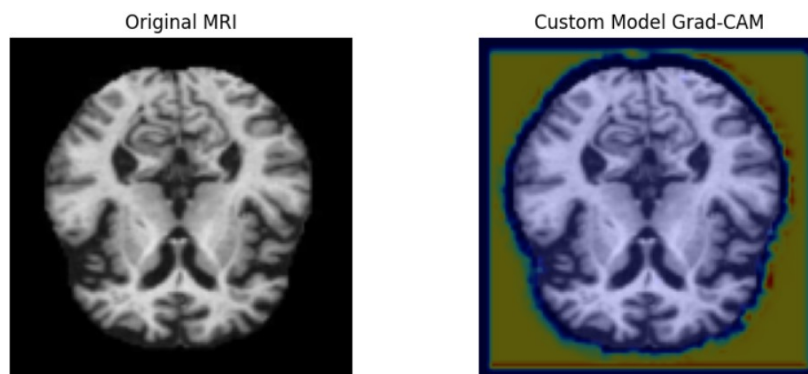


Figure 4.2: Grad-CAM shows which areas are most effective on the class prediction: blue mean low influence, red mean high influence

As can be seen from the right image, the network was extracting features such as: image edge (straight red line on the bottom), artifacts around the brain edge caused by MRI scanner. Thus the network was learning the background instead of clinically relevant brain features.

4.2 Improved Model Results and Performance

To improve the model, it was decided to realize the fine tuning of the hyperparameters using WandB sweeps, and the following configuration resulted in the best performance:

Parameter	Value
Best Validation Accuracy	0.5855
batch_size	64
input_shape	[3, 128, 128]
neurons_fc1	128
neurons_fc2	128
num_classes	4
dropout_rate	0.3
class_weights	[1.7864321, 24.802326, 0.5, 0.71433353]
learning_rate	$9.1155e^{-05}$
optimizer_name	adam

Table 4.1: Best Hyperparameters obtained with Wandb sweeps

The improved model yielded the following results:

Validate metric	DataLoader 0
val_acc	0.9559498429298401
val_f1	0.9591728448867798
val_loss	0.12116562575101852
val_prec	0.9590346813201904
val_rec	0.9596260786056519

Figure 4.3: Metrics of the improved model

The model obtained high accuracy and high recall, and the effectiveness of the model was confirmed by the Grad-CAM analysis as can be seen below:

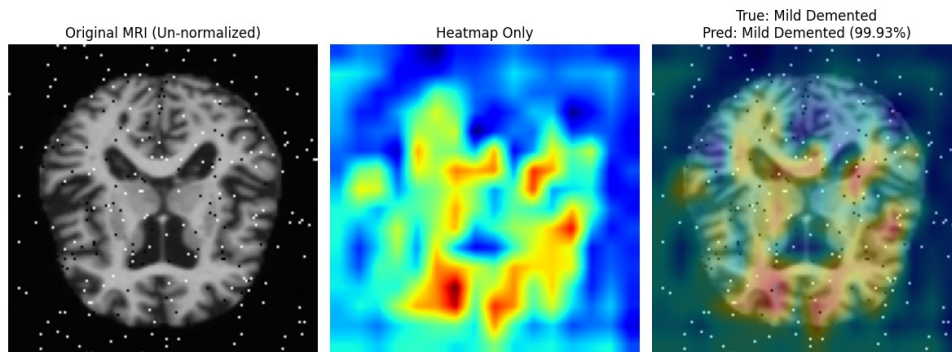


Figure 4.4: Grad-CAM result on the improved model

As can be seen, the class score was affected by the brain features that actually define if the person has either a healthy or an ill brain. Those results prove the obtained network was capable of satisfactory generalization and classifies information well.

It is generally preferable to maximize the Recall compared to minimize False Negatives. Missing a positive case (classifying a sick patient as healthy) is critical as it prevents early

treatment. Conversely, a False Positive is considered a safer error, as it would simply lead to further clinical examination by a doctor who can rule out the disease. Therefore, the model's high performance is particularly valuable as a preliminary screening tool.

Due to dataset adjustment, it was possible to effectively use Validation Accuracy as the target metric for the hyperparameter sweep without the risk of the model biasing towards the majority class. While Accuracy guided the optimization process, the final clinical reliability of the model was confirmed by its high Recall (Sensitivity), ensuring that positive cases are correctly identified.

4.3 Transfer Learning Model Results and Performance

After the improved model it was tried to implement a transfer learning approach using the ResNet18 architecture. The metrics obtained freezing the weight of the feature extractor, on the K-folds, as shown below, are not really high. While the results obtained when fine-tuning the entire network, without freezing the weights, are very high, comparable with the one obtained with the improved model.

Validate metric	DataLoader 0
val_acc	0.7483778433799744
val_rec	0.7561904191978825

Validate metric	DataLoader 0
val_acc	0.9446788430213928
val_f1	0.946605384349823
val_loss	0.1943112313747486
val_prec	0.957938551982771
val_rec	0.9365581403808594

Figure 4.5: In the left side are shown the metrics obtained tuning only the weights of the fully connected layers and on the right side the one obtained tuning all the weights of the ResNet18

Even if it was obtained a high accuracy and a high recall, with the overall weights tuning, the Grad-CAM doesn't give a good result as well as in the case of the improved model:

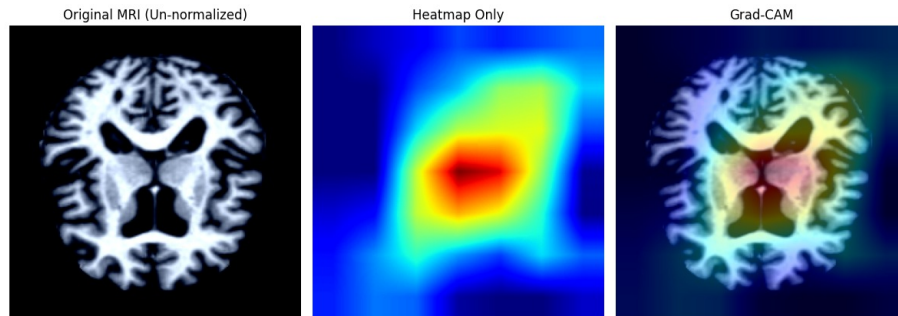


Figure 4.6: Grad-CAM shows which areas are most effective on the class prediction: blue -> low influence, red -> high influence

The visualization shows that the model correctly ignores the background noise and focuses on the brain. However, the heatmap is quite broad and focuses mainly on one side. This suggests the model is looking at a general area rather than specific, fine-grained details.

Chapter 5

Conclusion

The main objective of this project was successfully achieved: to develop a robust Neural Network model capable of classifying the four stages of Alzheimer's disease. The improved model demonstrated high performance, reaching a peak **overall accuracy of 95.5%** and a **Recall (Sensitivity) of 95.6%**, confirming strong generalization capability with no evidence of over-fitting, established by the Grad-CAM result.

About the Transfer Learning approach with ResNet18 proved highly effective, achieving a K-fold validation accuracy of 97%. However, while the quantitative metrics are excellent, the qualitative analysis via Grad-CAM introduces a degree of uncertainty. The heatmaps, although correctly localized around the ventricles, appear coarse and lack fine-grained definition. This suggests that while the model is statistically robust, its focus on specific anatomical features may not be as precise as the accuracy score implies.

To ensure these results and enhance model performance, the following key strategies were implemented:

- **K-fold Cross-Validation:** Utilized to maximize the effective use of the limited available dataset.
- **Class Weighting:** Applied during training to address the dataset imbalance, ensuring the model did not bias towards the majority class.
- **Batch Normalization and Leaky-ReLU:** Integrated into the architecture to aid the stability and speed of convergence.
- **WandB Hyperparameter Sweeps:** Employed for efficient optimization of the model's configuration.

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